

Neighborhood-Aware Stochastic Generation: Enforcing Fiber Group Coherence in the Conditional Joint Gaussian Model to Generate Coherent/Continuous 3D FRP microstructures

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Introduction

The current research groups' developed pipeline modelling engine fits a joint Gaussian copula to the marginal distributions of fiber positions and out-of-plane tilt angles within a local depth window. At generation time, each fiber's path is grown by sequentially sampling from this copula but each fiber is treated independently. There is no mechanism that forces fibers that were physical neighbors in the CT scan (real sample) to remain neighbors in the synthetic volume.

The spatial coherence analysis (Delaunay neighborhood persistence) quantifies exactly this gap. Currently, the specific fiber-pair identities that constitute persistent neighbors in the real material are largely not reproduced. This is physically consequential. In a real CFRP laminate, manufacturing constraints (ply-bed friction, compaction, fiber-fiber interactions and other factors) prevent fibers from drifting past one another. Groups of fibers move coherently through the thickness. A model that ignores this constraint will misrepresent the spatial stress-concentration topology relevant to delamination and crack propagation or manufacturing related analyses.

Aim: Extend the Sliding-Window Gaussian copula engine with a neighborhood-aware generation layer that (a) learns persistent fiber-group structure from CT data using a spatial graph model, and (b) enforces group cohesion during stochastic sampling without degrading the per-fiber marginal statistical fidelity.

Research Question: Can a graph-conditioned extension of the Sliding-Window Gaussian copula model increase the neighborhood stability ratio and coherent-group-count fidelity of synthetic CFRP microstructures, relative to the current

baseline, while preserving per-fiber misalignment angle and inter-slice copula fidelity ?

Activities & Expectations

1. Neighborhood characterization study: Using existing CT data and the Delaunay persistence code already in the repository, characterize fiber group structure: cluster size distributions, within-group vs between-group copula ρ . Determine whether groups follow a stationary spatial process or show long-range ordering.
2. Group assignment model: Develop a method to assign incoming seed-layer fibers to neighborhood groups. Candidates: K-means on centroid positions, community detection on the Delaunay graph etc.
3. Group-conditioned copula sampler: Modify the current engine so that within-group fiber pairs are sampled from a learned '*within-group*' fibers copula (higher ρ , tighter positional coupling)
4. Bayesian Optimization re-calibration: Extend the BO parameter space to include group-copula hyperparameters (within-group ρ scale, group-radius parameter).
5. Validation: Compare against the baseline and on: neighborhood stability ratio, coherent-group count, per-fiber θ_x/θ_y etc.

Expected outcome: A quantified improvement in spatial coherence metrics with documented trade-offs. Even a partial improvement (e.g. 20–30% increase in stability ratio) would constitute a meaningful contribution since it directly narrows the gap between the stochastic model and physical reality.

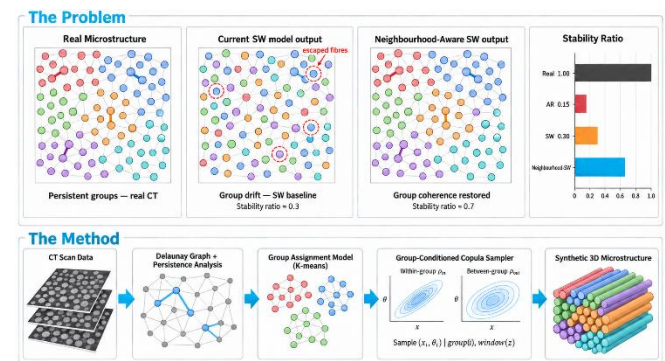


Figure 1. Proposed pipeline schematic [fig made by ChatGPT].

This work is suitable for Master/Bachelor students.
Sounds interesting? Please get in touch!

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